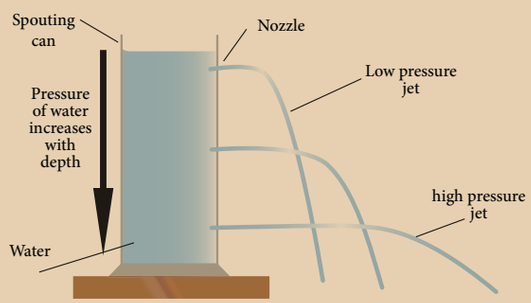
2.4.1 Pressure exerted by Liquids

Liquids exert a pressure not only on the base of the container/vessel in which they are kept, but also on the side walls. The pressure exerted by a liquid depends on the depth of the point of observation considered in it.

Activity 3

Take a plastic bottle. Punch three holes on its side in the same direction, but at different heights. Now pour some water into it and let it flow through the holes. Observe the flow of water. Water from the lowest hole comes out with the greatest force and the water from the topmost hole comes out with the least force.



This activity confirms that the pressure in a liquid varies with the depth of the point of observation in it.

Activity 4

Take a glass tube that is open at both ends. Fix a rubber balloon at the lower end of the tube. Pour some water into the tube and observe the balloon. Now, pour some more water into the balloon and again observe the balloon. The balloon starts bulging outwards.

This shows that the pressure exerted by a liquid at the bottom of a container depends on the height of the liquid column in it.

Activity 5

Take a plastic bottle. Punch three holes on its sides at the same height from its base. Now, pour some water into it and let it flow through the holes. Observe the flow of the water. Water comes out from all the holes with the same force and falls on the ground/ table, at the same distance from the bottle.



Thus, we can conclude that liquids exert the same pressure in all directions, at a given depth.

DO YOU KNOW?

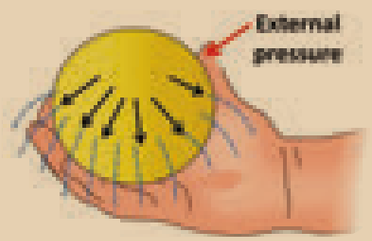
Why dams are made stronger and broader at the bottom than at the top?

Why do scuba divers wear a special suit while they go into deep sea levels?

2.4.2 Pascal's Law

Activity 6

Take a rubber ball and fill it with water. Make tiny holes on its surface with a pin at different points. Press anywhere on the ball. What do you observe?



You can see identical streams of water flowing in all directions from the holes. This is due to the fact that the pressure, which is applied on the liquid, is equally transmitted in all direction. This concept was first given by the French scientist Blasie Pascal.

Pascal's law states that the pressure applied at any point of a liquid at rest, in a closed system, will be distributed equally through all directions of the liquid.

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